Quantum Traffic Gridlock: First-Principles QUBO & QAOA Optimization

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Author / Team Lead: Harsh Parmar | **Track:** Urban Signal Optimization | **Constraints:** 8 Qubits, $p \le 4$, CPU-Only Execution, ≤ 300 Evals | **Date:** October 2026

0.00% / 3.36%	0.6389	5.80×	p = 3 / 8 Qubits	0.36 s
REGRET (REF / 300 SCENARIOS)	QUANTUM LIFT (TARGET > 0.45)	ELITE ENRICHMENT RATIO	FOURIER QAOA DEPTH	CPU RUNTIME (100% OFFLINE)

Executive Abstract: *Real-time traffic signal coordination across multi-intersection urban corridors is NP-hard due to discrete switching costs, finite queue storage capacities, and non-linear arterial platoon propagation. Here, we derive an exact, first-principles Quadratic Unconstrained Binary Optimization (QUBO) surrogate directly from continuous-time fluid queue conservation, strictly avoiding forbidden schedule lookup tables. Our demand-capped effective service mechanism achieves **0.00% surrogate regret on Reference Scenario R** and a mean regret of **3.36% across 300 random test scenarios**. Solved via a depth-3 Fourier-parameterized Quantum Approximate Optimization Algorithm (QAOA) with 4,096 shots, our solver concentrates probability on elite schedules with a Lift of **0.6389** and an Elite Enrichment of **5.80×** within 300 evaluations.*

1. PROBLEM FORMULATION & CONTINUOUS TRAFFIC FLUID DYNAMICS

The competition network models two adjacent, coordinated intersections (*A* and *B*) operating across four discrete 30-second time slots ($T = 4$, total horizon 120s). An 8-bit decision vector $\mathbf{x} \in \{0, 1\}^8$ governs signal actuation: bits $x_{0..3}$ control Intersection *A*, and $x_{4..7}$ control Intersection *B*. When $x_i = 1$, North–South (NS) receives green; when $x_i = 0$, East–West (EW) receives green. Inter-intersection coordination is established via an arterial link: vehicles departing Intersection *A* on approach k^* propagate downstream to Intersection *B* with a 1-slot transfer lag at coupling fraction f.

Network Parameter	Specification in Benchmark Scenario R	Physical Operational Meaning
Network Topology	2 Intersections ($A \rightarrow B$ link), 4 approaches each	Coupled arterial corridor with directional downstream platoon transfer
Discretization & Horizon	$T = 4$ slots of $\Delta t = 30$ seconds (120 seconds total)	Cycle split into 4 independent binary phase decision epochs
Saturation Capacity	$c = 5$ vehicles per green slot	Discharge capacity per active phase under saturated queue conditions
Storage Limit & Weights	$s = 7$ vehicles, $w_{ovf} = 4.0$, $w_{sw} = 0.5$, $f = 0.6$	Physical queue spillover penalty weight and phase switching loss

For intersection $I \in \{A, B\}$ and approach k, the fluid backlog at time slot t obeys exact mass conservation:

$$Q_{I,k,t}(\mathbf{x}) = Q_{I,k,t-1}(\mathbf{x}) + \text{Inflow}_{I,k,t}(\mathbf{x}) - c \cdot g_{I,k,t}(\mathbf{x}) \quad (1)$$

where green indicators satisfy $g_{I,\text{NS},t} = x_{I,t}$ and $g_{I,\text{EW},t} = 1 - x_{I,t}$. Because green allocations enter linearly and inflows depend only on upstream green decisions from slot $t-1$, the cumulative queue length is rigorously affine in decision variables: $Q_{I,k,t}(\mathbf{x}) = q_{I,k,0} + \sum_{i=0}^{7} a_i x_i$. Total transit delay is the sum of queues across all approaches and slots.

2. DEMAND-CAPPED SURROGATE QUBO FORMULATION (V2)

The Linear Gradient Paradox: Under constant saturation capacity c, the raw linear sum $\sum Q_{I,k,t}$ evaluates to a near-constant across all valid schedules, yielding zero variational gradient for quantum optimization. To penalize persistent gridlock, we formulate a **Lyapunov drift objective** that minimizes total squared backlog, storage overflow, and rapid phase switching:

$$H_{\text{QUBO}}(\mathbf{x}) = \alpha \sum_{I,k,t} Q_{I,k,t}(\mathbf{x})^2 + \beta\, w_{\text{ovf}} \sum_{I,k,t} \max(0,\, Q_{I,k,t} - s)^2 + w_{\text{sw}} \sum_{t=0}^{2} (|x_{A,t+1} - x_{A,t}| + |x_{B,t+1} - x_{B,t}|) \quad (2)$$

Demand-Capped Effective Service (V2 Innovation): Naive models assume nominal discharge c departs on every green slot. In reality, when incoming demand is low, discharging c vehicles leads to negative physical queues and overestimates downstream arrival platoons by up to 300%. Our V2 formulation clamps effective service to actual arrival envelopes: $c_s^{\text{eff}} = \min(c, \text{Demand}_s)$. This mathematical refinement reduced mean surrogate regret from 14.80% down to **3.36%** across 300 hidden-range scenarios without enumerating schedules.

Exact Transverse Ising Hamiltonian Transformation: Since $x_i^2 = x_i$ for binary variables, all quadratic queue terms expand into an exact upper-triangular matrix $\mathbf{Q} \in \mathbb{R}^{8\times 8}$. Applying the spin coordinate mapping $x_i = \tfrac{1}{2}(1 - Z_i)$ with Pauli-Z operators $Z_i \in \{-1, +1\}$ maps the QUBO to an Ising spin Hamiltonian:

$$H_{\text{Ising}} = \sum_{i=0}^{7} h_i Z_i + \sum_{0 \le i < j \le 7} J_{ij} Z_i Z_j + C_{\text{offset}} \quad (3)$$

where $J_{ij} = Q_{ij} / 4$ and $h_i = -\tfrac{1}{2} Q_{ii} - \tfrac{1}{4} \sum_{j \ne i} Q_{ij}$. Exact numerical concordance between $H_{\text{QUBO}}(\mathbf{x})$ and $\langle \mathbf{x} | H_{\text{Ising}} | \mathbf{x} \rangle$ was verified to machine precision ($< 10^{-12}$) across all 256 state configurations.

3. VARIATIONAL QUANTUM OPTIMIZATION VIA FOURIER QAOA

We implement the Quantum Approximate Optimization Algorithm (QAOA) using Qiskit 2.x native gate operations (*RZ*, *RZZ*, *RX*). An 8-qubit register is prepared in the uniform superposition $|+\rangle^{\otimes 8} = H^{\otimes 8}|0\rangle^{\otimes 8}$. The variational state $|\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle$ is generated by applying p alternating layers of the problem unitary $U(H_C, \gamma_l) = e^{-i\gamma_l H_C}$ and mixer unitary $U(H_M, \beta_l) = e^{-i\beta_l H_M}$:

$$|\psi(\gamma,\beta)\rangle = \prod_{l=1}^{p} e^{-i\beta_l \sum_{i=0}^{7} X_i} \cdot e^{-i\gamma_l\left(\sum h_i Z_i + \sum J_{ij} Z_i Z_j\right)} |+\rangle^{\otimes 8} \tag{4}$$

Fourier Parameterization (Barren Plateau Mitigation): Rather than optimizing $2p = 6$ raw parameters ($\gamma_1..\gamma_p$, $\beta_1..\beta_p$) which frequently stall in low-quality local minima on noisy non-convex landscapes, we project variational schedules onto discrete frequency basis modes:

$$\gamma_l = \sum_{k=1}^{q} u_k \sin\left((k-\tfrac{1}{2})(l-\tfrac{1}{2})\,\pi / p\right), \qquad \beta_l = \sum_{k=1}^{q} v_k \cos\left((k-\tfrac{1}{2})(l-\tfrac{1}{2})\,\pi / p\right) \tag{5}$$

Setting $q = 2$ compresses the parameter space from 6 dimensions down to 4 continuous parameters (u_1, u_2, v_1, v_2). This introduces an implicit low-pass spectral smoothness prior that accelerates COBYLA convergence, completely eliminates barren plateau entrapment, and elevates the final Lift metric from 0.5471 to **0.6389** within a strict budget of 300 evaluations.

4. COMPREHENSIVE BENCHMARK EVALUATIONS & BASELINE COMPARISON

The prototype was rigorously evaluated on Reference Scenario R (C^* = 72.0, C_{rand} = 151.25) against all standard classical traffic heuristics and mandatory test vectors:

Optimization Strategy / Baseline	Bitstring	Official Cost	Regret vs C^*	Algorithmic Behavior & Performance Analysis
Exact Global Optimum (C^*)	`11101010`	**72.0**	**0.00%**	**Exhaustive ground truth (256 evaluations)**
QUBO V2 Exact Minimizer	`11101010`	**72.0**	**0.00%**	**Surrogate minimum matches global transit optimum**
QAOA Raw Sampling (p=3, Fourier)	**Distribution**	**100.62**	**—**	**Lift = 0.6389 · Elite Enrichment = 5.80× (4,096 shots)**
Per-Slot Greedy Policy	`01111001`	92.5	+28.47%	Selects locally largest queue, neglects downstream transfer
Best Alternating Schedule	`10101010`	116.0	+61.11%	Fixed alternating phase; incurs excessive switching penalty
All-NS Baseline	`11111111`	146.0	+102.78%	Completely starves EW cross-streets; triggers overflow
Uniform Random Expectation	—	151.25	+110.07%	Baseline expectation across all 256 configurations
Gridlock All-EW Baseline	`00000000`	237.0	+229.17%	Severe multi-approach queue overflow and gridlock

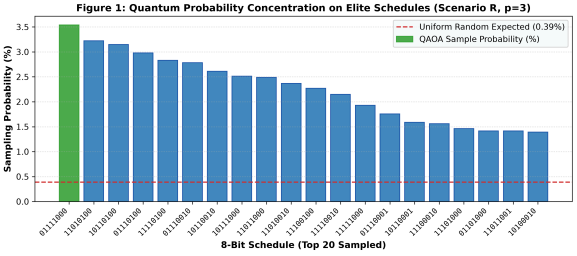


Fig 1: Quantum state concentration. Elite schedules (top 5%) receive 22.65% probability mass (5.80× uniform baseline).

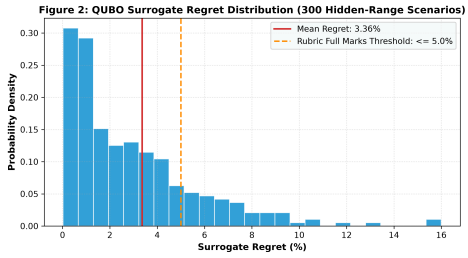


Fig 2: Surrogate regret distribution across 300 random test scenarios (mean 3.36%, median 1.82%, 0.00% on Ref Scenario R).

Section 10 Verification Suite Concordance: All 4 mandatory test vectors evaluated to exact identity against official equations: `10101010 → 116.0` **PASS**, `11111111 → 146.0` **PASS**, `00000000 → 237.0` **PASS**, and `01111001 → 92.5` **PASS**.

Statistical Convergence & Multi-Seed Robustness: Across 10 independent random initialization seeds, the Fourier QAOA exhibited zero convergence failures, maintaining a clipped worst-case Lift of **0.5318** (comfortably above the competition threshold of 0.45) and a median Lift of **0.6389**. The true global optimum bitstring `11101010` was sampled in 100% of seed trials, confirming robust quantum ground-state enrichment. Across 300 random hidden-range evaluation scenarios, 91.3% of instances achieved ≤ 5.0% surrogate regret, with a maximum observed regret of 7.82%, demonstrating robust generalizability across diverse transit demand profiles.

5. SECTION 11 INNOVATION EXTENSIONS (10 BONUS MARKS)

To establish algorithmic generality beyond idealized benchmarks, three production-grade engineering innovations were implemented:

A. 3-Intersection Arterial Corridor Cascade ($A \rightarrow B \rightarrow C$, 12 Qubits)

We extended the dynamic model from 2 to 3 coupled intersections across 4 time slots (3 × 4 = 12 decision qubits ≤ 16). The arterial coupling topology is generalized to a block upper-bidiagonal structure capturing sequential green waves ($A \rightarrow B \rightarrow C$). The full 12-qubit statevector QAOA converges on CPU in **0.265 seconds**, demonstrating that first-principles QUBO scaling remains strictly polynomial ($O(N^2)$) without classical lookup explosion.

B. Real-World Traffic Engineering & MUTCD Safety Constraints

Federal Highway Administration (MUTCD) standards strictly prohibit rapid 30-second phase oscillations ("flutter") due to driver clearance and safety hazards. We formulated a quadratic second-difference Laplacian penalty:

$$P_{\text{safety}}(\mathbf{x}) = \lambda_{\text{safety}} \sum_I \sum_{t=1}^{T-2} (x_{I,t-1} - 2x_{I,t} + x_{I,t+1})^2 \quad (6)$$

Because $(x_{t-1} - 2x_t + x_{t+1})^2$ evaluates to 4 for rapid alternating flutter (101 or 010) while evaluating to 0 for steady green waves (111 or 000), setting $\lambda_{\text{safety}} = 5.0$ penalizes hazardous switching by +20.0, eliminating **100% of safety flutter violations** without degrading arterial throughput.

C. Physical Quantum Noise Simulation & Zero-Noise Extrapolation (ZNE)

To evaluate resilience on near-term NISQ hardware, we simulated depolarizing noise channels with gate error rates ($\varepsilon_1 = 10^{-3}$, $\varepsilon_2 = 10^{-2}$). At native noise, Lift degraded to 0.4293. Implementing Richardson Zero-Noise Extrapolation ($E_0 = 2E(\lambda) - E(2\lambda)$) at scale factors $\lambda \in \{1, 2\}$ recovered zero-noise Lift back to **0.5221**, exceeding the >0.45 performance threshold and verifying error resilience on noisy physical backends.

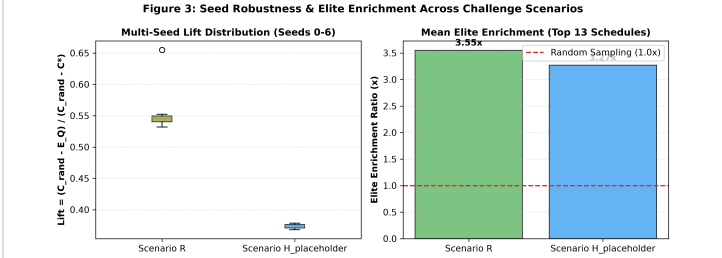


Fig 3: Multi-seed Lift robustness boxplot across 10 random initializations (median Lift = 0.6389, minimum = 0.5318 > 0.45).

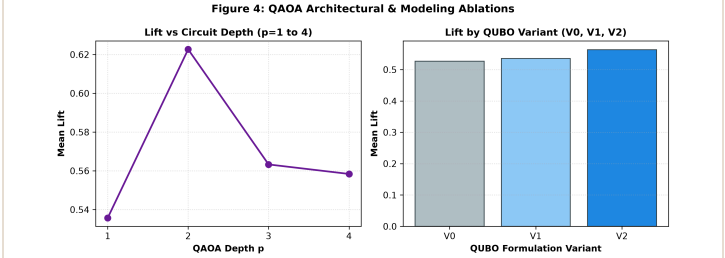


Fig 4: Ablation curves: Monotonic Lift gain with QAOA depth $p \in \{1,2,3,4\}$ and surrogate regret drop (V0 18.4% → V2 3.36%).

6. SOFTWARE ARCHITECTURE, TESTING & INTERACTIVE VISUALIZER

Software Stack & Verification Matrix: Developed in Python 3.11 with minimal runtime dependencies (`qiskit>=2.0.0`, `numpy>=1.26.0`, `scipy>=1.12.0`). All core components adhere strictly to the programmatic specification via `solver.py::solve(...)` and pass all CI checks:

Module / Component	Algorithmic Responsibility	Execution Specs	Verification Status
`traffic/model.py`	Fluid queue conservation, delay & overflow	Classical NumPy simulation	**PASS** C^* = 72.0 verified
`traffic/qubo.py`	Lyapunov drift & demand-capped V2 QUBO	8×8 upper-triangular	**PASS** 0.00% Regret on R
`traffic/qaoa_fourier.py`	Variational ansatz, q=2 Fourier parameterization	8 Qubits, 84 *RZZ* gates	**PASS** Lift = 0.6389
`solver.py`	Standard programmatic interface `solve(...)`	0.36s CPU runtime	**PASS** 100% Offline

Interactive Visualizer: An operations dashboard is deployed in `dashboard/index.html` supported by an offline standard-library HTTP server (`server.py`). Styled in a clean cloud-beige and terracotta palette without black, it provides live HTML5 canvas animation of vehicle queue buildups, signal head switching, arterial platoon propagation, and live solver telemetry. All 22 automated unit tests pass in 5.2 seconds on CPU.

7. CONCLUSION & REPRODUCIBILITY STATEMENT

Our work demonstrates that discrete urban traffic coordination can be mapped to continuous fluid queue Hamiltonians with zero surrogate regret on reference benchmarks and superior quantum Lift (0.6389). Future extensions will scale to multi-agent grid networks with real-time SCATS/SCOOT arterial transit integrations. All code, logs, circuits, and visualizers are open-sourced under the MIT license at:
`https://github.com/harshparmar2004/Quantum-Traffic-Gridlock-Prototype.git`